

V2I-BASED SMART TRAFFIC SIGNAL SYSTEM WITH ADAPTIVE TIMING AND EMERGENCY VEHICLE PREEMPTION

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PROBLEM STATEMENT

Existing traffic signal systems operate on fixed timing and cannot adapt to real-time traffic conditions, leading to traffic congestion, increased waiting time, fuel wastage, and delays for emergency vehicles. They also do not predict the arrival of approaching vehicles, resulting in inefficient signal allocation. This project proposes a V2I-Based Smart Traffic Signal System with Adaptive Timing, Predictive Green Time, and Emergency Vehicle Preemption, where traffic signals dynamically adjust green time based on current traffic density and estimated vehicle arrival time. The system also provides immediate priority to emergency vehicles, ensuring smoother traffic flow, reduced congestion, and faster emergency response.

ABSTRACT

Rapid urbanization and the continuous increase in the number of vehicles have made traffic congestion one of the major challenges in modern transportation systems. Conventional traffic signals operate using fixed-time control, which cannot respond to changing traffic conditions. This results in unnecessary waiting at intersections, increased fuel consumption, higher vehicle emissions, and inefficient utilization of road infrastructure. Moreover, emergency vehicles such as ambulances, fire trucks, and police vehicles often experience delays at traffic signals, reducing the effectiveness of emergency response and potentially endangering lives. Therefore, an

intelligent traffic management system capable of adapting to real-time traffic conditions and prioritizing emergency vehicles is essential.

This project proposes a V2I-Based Smart Traffic Signal System with Adaptive Timing, Predictive Green Time, and Emergency Vehicle Preemption. The system enables wireless communication between vehicles and the traffic signal controller using the nRF24L01 transceiver. Each vehicle continuously transmits information such as vehicle type (normal or emergency), speed, and approach status. An ultrasonic sensor installed at the traffic signal measures the queue length to estimate the current traffic density. The controller processes both the queue length and the estimated arrival time of approaching vehicles to predict traffic demand and dynamically determine the optimal green signal duration. Unlike conventional adaptive systems that only respond to existing traffic queues, the proposed system predicts incoming traffic and adjusts the signal proactively, reducing congestion before it builds up.

In the event of an approaching emergency vehicle, the system immediately interrupts the normal signal sequence and provides a green signal to the corresponding lane while keeping all other directions red. After the emergency vehicle safely crosses the intersection, the controller automatically restores the normal adaptive traffic operation. This real-time priority mechanism significantly reduces emergency response delays while maintaining efficient traffic flow for other vehicles.

The proposed system is implemented using low-cost embedded hardware, including an STM32/Arduino controller, nRF24L01 wireless communication modules, ultrasonic sensors, and traffic signal LEDs, making it economical, scalable, and suitable for academic demonstrations and smart city research. Compared to existing systems that separately implement adaptive traffic control or emergency vehicle preemption, this project integrates adaptive signal timing, predictive vehicle arrival estimation, and emergency priority control into a single intelligent platform.

OBJECTIVES

- 1.To design and develop a V2I-based smart traffic signal system using wireless communication between vehicles and traffic infrastructure.
- 2.To dynamically adjust traffic signal timing based on real-time traffic density using an ultrasonic sensor.
- 3.To implement predictive green time by estimating the arrival time of approaching vehicles using their speed and approach information.
- 4.To provide automatic emergency vehicle preemption, allowing ambulances, fire trucks, and police vehicles to receive immediate signal priority.
- 5.To reduce vehicle waiting time, traffic congestion, fuel consumption, and carbon emissions through intelligent traffic management.
- 6.To improve emergency response time and enhance road safety using a low-cost, real-time embedded system prototype
- 7.To demonstrate a scalable and cost-effective solution suitable for future smart city transportation systems.
- 8.To integrate adaptive timing, predictive traffic control, and emergency vehicle prioritization into a single system.
- 9.To provide a reliable prototype suitable for future smart city traffic management applications.
- 10.To reduce traffic congestion at busy intersections.

SCOPE OF THE PROJECT

The scope of this project is to design, develop, and implement a V2I-Based Smart Traffic Signal System with Adaptive Timing, Predictive Green Time, and Emergency Vehicle Preemption to improve traffic management at road intersections. The system focuses on establishing real-time wireless communication between vehicles and traffic infrastructure using nRF24L01 transceivers. It continuously receives information such as vehicle type, speed, and approach status from vehicles while simultaneously monitoring traffic density using an ultrasonic sensor. Based on this information, the controller dynamically adjusts the traffic signal timing according to the current queue length and predicts the arrival of approaching vehicles to allocate the optimum green signal duration. This predictive approach helps reduce congestion before it develops, resulting in smoother traffic flow and improved intersection efficiency. The project also incorporates an emergency vehicle preemption mechanism that automatically detects ambulances, fire trucks, and police vehicles and immediately overrides the normal signal sequence to provide a dedicated green signal for safe and rapid passage. After the emergency vehicle crosses the intersection, the system restores the normal adaptive traffic control process without affecting overall traffic efficiency. The proposed prototype is implemented using cost-effective embedded hardware such as STM32, nRF24L01 wireless modules, ultrasonic sensors, and traffic signal LEDs, making it suitable for educational, research, and demonstration purposes. Furthermore, the project provides a scalable architecture that can be expanded from a single intersection to multiple interconnected intersections. It also serves as a foundation for future enhancements, including GPS-based vehicle tracking, AI and machine learning for traffic prediction, cloud-based traffic monitoring, IoT integration, and advanced V2X communication technologies. The proposed system contributes toward the development of intelligent transportation systems by reducing vehicle waiting time, minimizing fuel consumption and carbon emissions, improving emergency response efficiency, enhancing road safety, and supporting future smart city traffic management infrastructure.

EXPECTED OUTCOME

1. Develop a functional V2I-based smart traffic signal prototype using embedded hardware.
2. Dynamically adjust traffic signal timing based on real-time traffic density
3. Predict vehicle arrival and allocate optimum green signal duration
4. Reduce vehicle waiting time at road intersections.
5. Minimize traffic congestion and improve overall traffic flow.
6. Provide automatic priority to emergency vehicles through signal preemption.
7. Reduce emergency vehicle response time and ensure faster passage.
8. Lower fuel consumption and vehicle emissions by reducing idle time.
9. Demonstrate reliable wireless communication between vehicles and traffic signals using nRF24L01
10. Provide a scalable and cost-effective solution that can be extended for future smart city and intelligent transportation systems.
11. Improve road safety by reducing traffic conflicts and ensuring orderly vehicle movement at intersections.
12. Demonstrate the feasibility of integrating adaptive timing, predictive green time, and emergency vehicle preemption into a single intelligent traffic management system

LITERATURE SURVEY

S.NO	PAPER TITLE	CONTRIBUTION	LIMITATIONS
1	Adaptive Traffic Signal Control Based on Traffic Density	Adjusts signal timing according to vehicle density to reduce congestion	Does not support emergency vehicle priority.
2	Vehicle-to-Infrastructure (V2I) Communication for Intelligent Transportation Systems	Enables real-time communication between vehicles and traffic infrastructure.	Focuses mainly on communication without adaptive traffic control
3	Emergency Vehicle Preemption at Signalized Intersections	Provides automatic green signals for emergency vehicles.	Does not optimize signal timing for normal traffic
4	IoT-Based Smart Traffic Monitoring and Control System	Uses IoT sensors for traffic monitoring and signal control.	No predictive traffic analysis or V2I communication
5	Intelligent Traffic Signal Control Using Wireless Sensor Networks	Uses wireless sensors to detect traffic density and optimize signal timing	Limited communication with moving vehicles.
6	Predictive Traffic Signal Control Using Vehicle Arrival Estimation	Predicts vehicle arrival time to allocate green signal efficiently	Does not handle emergency vehicle prioritization.

LITERATURE SURVEY

S.NO	PAPER TITLE	CONTRIBUTION	LIMITATIONS
7	AI-Based Traffic Signal Optimization for Smart Cities	Uses artificial intelligence to optimize traffic signal timing	Requires high computational resources and large datasets
8	Smart Traffic Management Using Embedded Systems	Implements embedded controllers for automated traffic management	Limited scalability and lacks V2I communication.
9	Real-Time Traffic Signal Control Using Ultrasonic Sensors	Measures vehicle queue length using ultrasonic sensors for adaptive timing.	Reacts only to current traffic and cannot predict incoming vehicles.
10	Deep Learning-Based Adaptive Traffic Signal Control for Smart Cities	Uses deep learning to analyze traffic patterns and optimize signal timing in real time, improving traffic flow and reducing congestion	Requires large amounts of training data, high computational resources, and does not provide dedicated emergency vehicle preemption through V2I communication.

EXISTING SOLUTION ANALYSIS

S.NO	EXISTING SOLUTION	FEATURES	LIMITATIONS
1	Fixed-Time Traffic Signal System	Operates using predefined signal timings regardless of traffic conditions.	Cannot adapt to real-time traffic, causing congestion and unnecessary delays.
2	Traffic Density-Based Signal System	Adjusts signal timing based on the number of waiting vehicles using sensors.	Reacts only after congestion forms and cannot predict approaching traffic
3	Emergency Vehicle Preemption System	Provides signal priority for emergency vehicles using GPS, RFID, or siren detection.	Focuses only on emergency vehicles and does not optimize normal traffic flow.
4	V2I-Based Traffic Management System	Enables communication between vehicles and traffic infrastructure for real-time information exchange.	Requires costly communication technologies and lacks predictive traffic management
5	AI/IoT Smart Traffic Management System	Uses IoT devices, cameras, and AI algorithms to monitor and optimize traffic	High implementation cost, complex infrastructure, and significant computational requirements.

GAP ANALYSIS

Lack of Predictive Traffic Management

Most existing traffic signal systems respond only to current traffic conditions and do not predict the arrival of approaching vehicles, leading to inefficient signal timing.

Separate Emergency Vehicle Priority

Emergency vehicle preemption is usually implemented as an independent system and is not integrated with adaptive traffic signal control, reducing overall efficiency.

High Implementation Cost

Many V2I-based traffic management systems rely on expensive technologies such as DSRC, Cellular-V2X, cameras, and advanced sensors, making them difficult to deploy on a large scale.

Inefficient Traffic Flow

Existing adaptive systems mainly adjust signal timing based on queue length after congestion has already formed, instead of preventing congestion proactively.

Limited Integration

Most current solutions focus on a single feature such as adaptive timing, emergency vehicle priority, or V2I communication, rather than combining all these functionalities into one intelligent system.

Lack of Cost-Effective Solutions

There are very few low-cost embedded system prototypes that integrate V2I communication, predictive green time, and emergency vehicle preemption, making practical implementation difficult for educational and small-scale smart city applications.

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